Server Setup

Enrique Calderon

LIDSoL

Semester 2026-2

Server Setup Overview

Goals

Attendees configure a VPS or VM with SSH hardening basics, firewall rules, Podman installation, and a systemd/quadlet service.

Duration

90 minutes

Connecting to the Server

Connect to your target machine using SSH.

```
ssh user@your.vms.ip
```

Securing the Server: SSH Hardening

- Disable root login and password authentication (use Keys).
- Edit `/etc/ssh/sshd_config`:

```
PermitRootLogin no
PasswordAuthentication no
```

- Restart `sshd` to apply changes.

Firewall Rules (Firewalld)

- Allow only what you need.
- Installs`firewalld`:

```
sudo dnf install firewalld
sudo systemctl enable --now firewalld

sudo firewall-cmd --set-default-zone=public --permanent
sudo firewall-cmd --add-service=ssh --permanent
sudo firewall-cmd --add-service=http --permanent
sudo firewall-cmd --add-service=https --permanent
sudo firewall-cmd --reload
```

Podman Installation

- Install podman on the target server.
- Example for fedora-based distros:

```
sudo dnf update
sudo dnf install podman
```

- Ensure quadlet features are supported (Podman $>$ v4.4).

Deploying via Systemd/Quadlet

- **Quadlet** allows writing simple systemd service files to manage Podman containers automatically.
- Let systemd define the lifecycle natively.

Creating the Quadlet file

In your User Systemd directory (`~/.config/containers/systemd/`):

`my-flask-app.container`

```
[Unit]
Description=My Flask Backend App

[Container]
Image=my-flask-app:latest
PublishPort=5000:5000
EnvironmentFile=/home/user/app/.env

[Install]
WantedBy=default.target
```

Enable with Systemd

Reload user systemctl daemon to read the Quadlet file, which auto-generates a service:

```
systemctl --user daemon-reload
systemctl --user start my-flask-app
```

Check the status:

```
systemctl --user status my-flask-app
```

Try accessing the service from the server:

```
curl http://127.0.0.1:5000/ -X POST \
  -H "Content-Type: application/json" \
  -d '{"secret_key": "supersecretkey"}'
```